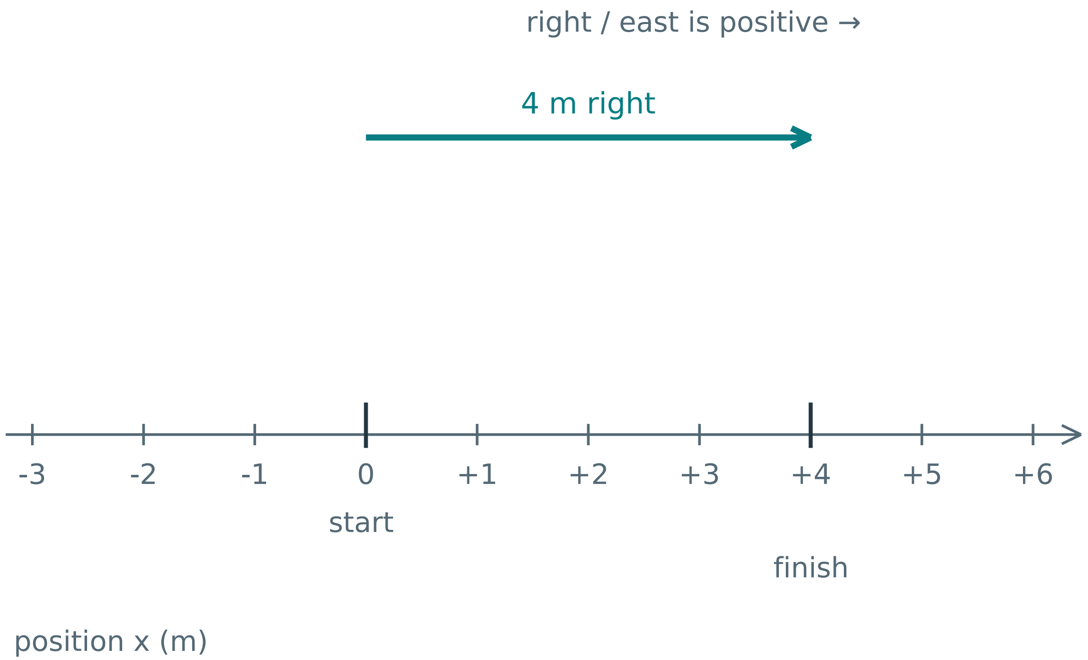
First, find your position on the line

Imagine a straight corridor. We choose a door as 0 m and right as positive.



The numbers are positions measured from the door. They are not distances already walked.

What does +4 m tell us?

It says where you are: 4 m to the right of the door. A position of −2 m means 2 m to its left.

The symbol *x* means position. The sign tells us which side of our chosen zero we mean.

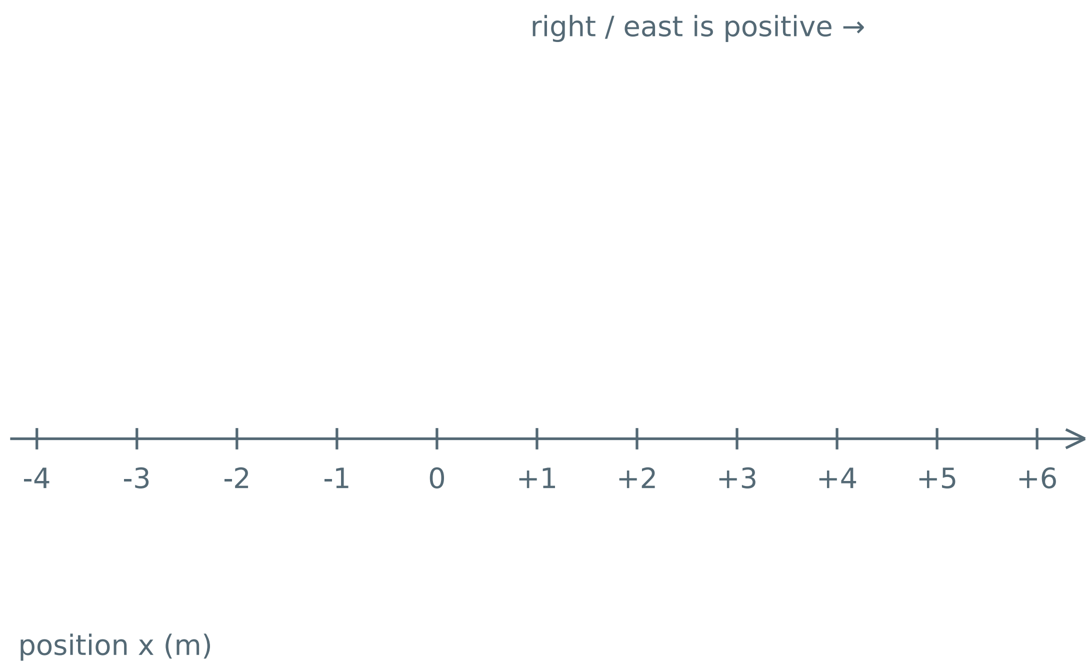
Move your finger

Trace from 0 to +4. Then trace back to 0. The second move goes left, even though every point along it is at 0 m or above.

Say it aloud: position tells me where I am. The direction of my movement tells me which way I am going.

Your turn · Do one small step at a time

Use the empty number line. Mark your answers before checking them.



Use different marks for a position and a movement arrow.

G1a · Locate a position

Mark −2 m. Is this left or right of the origin?

G1b · Make one movement

Start at +1 m. Draw an arrow 3 m left. Label where it ends.

G1c · Compare endpoints

A different move starts at −2 m and finishes at +3 m. The arrow points right. Complete: change in position = ___ m right.

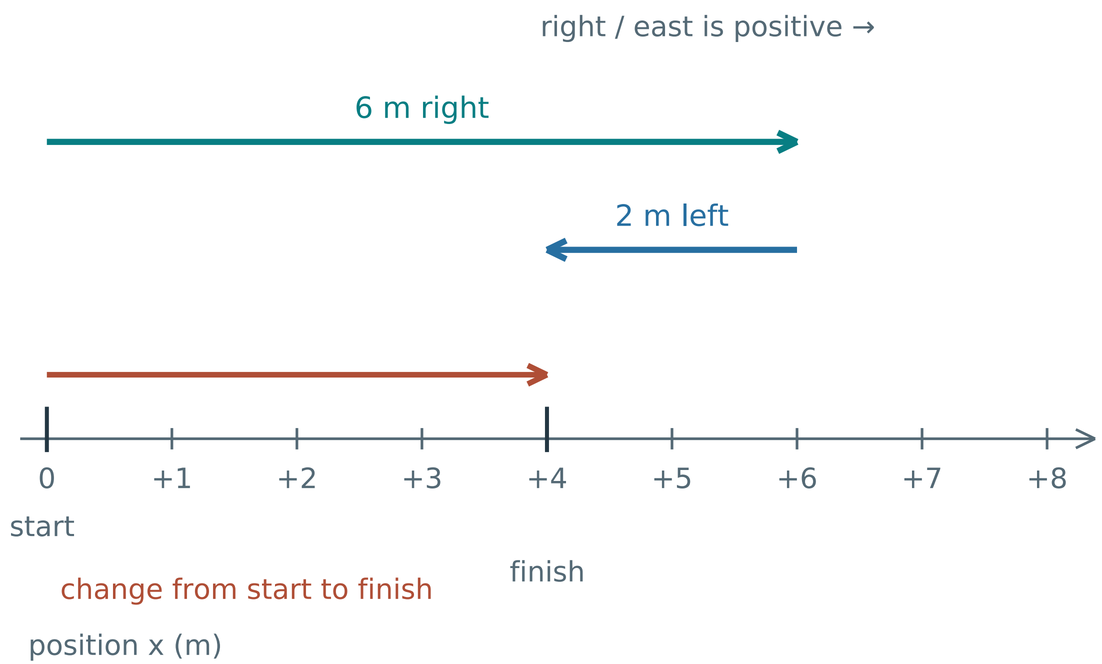
Say which number describes a location and which number describes a change. Check the small-step solutions only after you have marked the line.

Check G1 · p. 17

1 · SEE THE WHOLE PATH AND THE ENDPOINTS

One journey, two useful questions

Start at 0 m, walk 6 m right, then walk 2 m left.



The blue return arrow is still part of the journey. The red arrow joins only start and finish.

How much ground did you cover?

Your feet travelled both parts. Add their lengths.

Distance = 6 + 2 = 8 m

What changed from start to finish?

You finish 4 m to the right of where you started. That change is displacement.

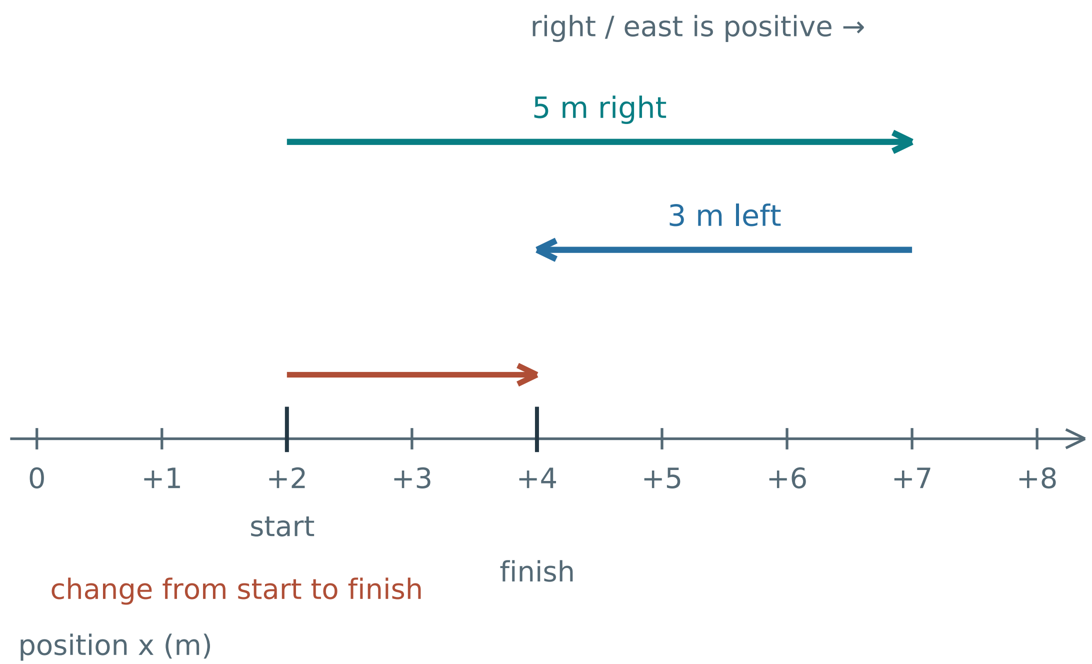
Displacement = +4 m

Distance has no direction. In a straight line, the sign of displacement gives its direction.

Keep the two questions separate: “How much path?” → distance. “What change from the start?” → displacement.

Your starting position matters

Start at +2 m, go to +7 m, then return to +4 m.



The finish is +4 m, but the change from +2 m to +4 m is only +2 m.

Read the path first

Distance = 5 + 3 = 8 m

Now compare the endpoints

A change is the new value minus the old value. Here, final position minus initial position gives:

Displacement = +4 − (+2)

= +2 m (right)

A useful check: the size of the displacement cannot exceed the total path length.

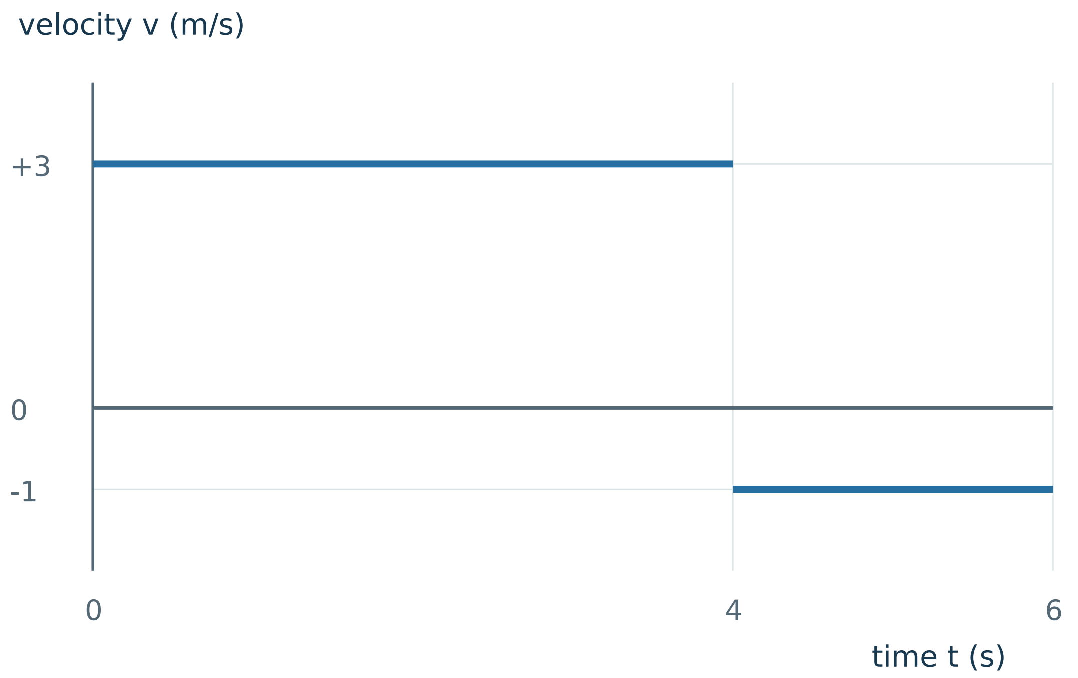
If you return to your start, the endpoint change is zero. Your walking is not erased: distance still counts every part. Try A1–A4 next.

Try A1 · p. 10

Try A3 · p. 11

A velocity–time graph tells a story

Choose right as positive. Read the graph from left to right as time passes.



The horizontal coordinate is time, not position. The line is not the shape of a road.

First four seconds

The line stays at +3 m/s. This means the object moves 3 m right in each second.

Next two seconds

The line is at −1 m/s. The object now moves 1 m left in each second. Its speed is 1 m/s.

This interval lasts 6 − 4 = 2 s. Always subtract the two time readings.

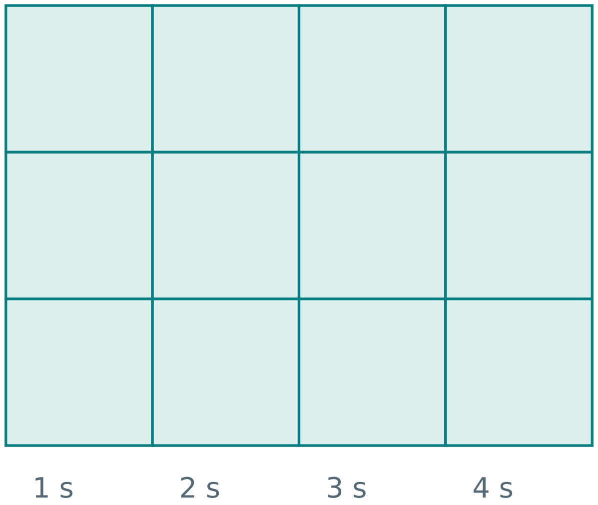
The abrupt switch is an idealisation. It contributes no extra time or distance.

A horizontal line above zero means steady motion. A line on v = 0 means rest. Below zero means motion in the chosen negative direction.

Why does graph area give a change in position?

At +3 m/s, each second adds another 3 m of rightward movement.

Each cell represents 1 m



Each column: 3 m travelled in 1 s

Four one-second columns make 12 m. This is the +3 m/s part of the previous graph.

Count what happens each second

$3 + 3 + 3 + 3 = 12$ m

Adding the same change four times is multiplication.

Change = velocity × duration

On a v–t graph, height is velocity and width is duration. Their product is the rectangle area.

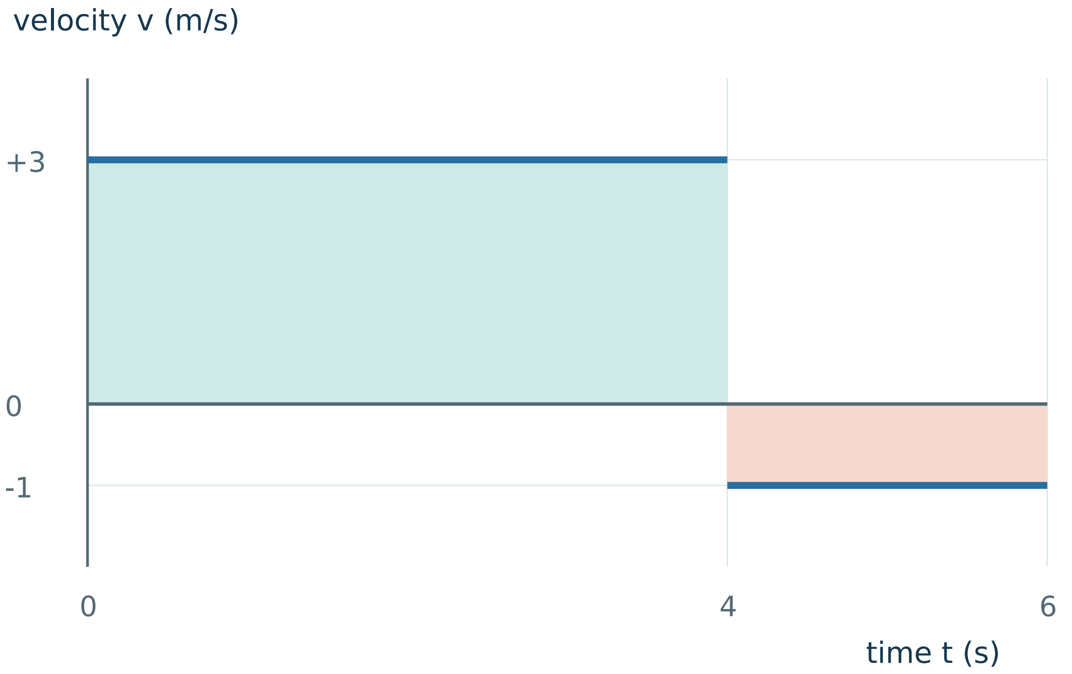
$(+3\ \text{m/s}) \times (4\ \text{s}) = +12\ \text{m}$

The seconds cancel, leaving metres. This rectangle rule requires constant velocity on the interval.

Do not use the final velocity for an entire interval when velocity changes. In that case, use the actual area under the graph.

A return journey counts in two different ways

Use the same graph: +3 m/s for 4 s, then −1 m/s for 2 s.



Above zero adds rightward change. Below zero adds leftward change. Neither part is “negative walking”.

First find each change

Rightward part: $(+3) \times 4 = +12$ m.
Leftward part: $(-1) \times 2 = -2$ m.

Displacement: signed changes

$$12 + (-2) = +10 \text{ m}$$

Count all the movement

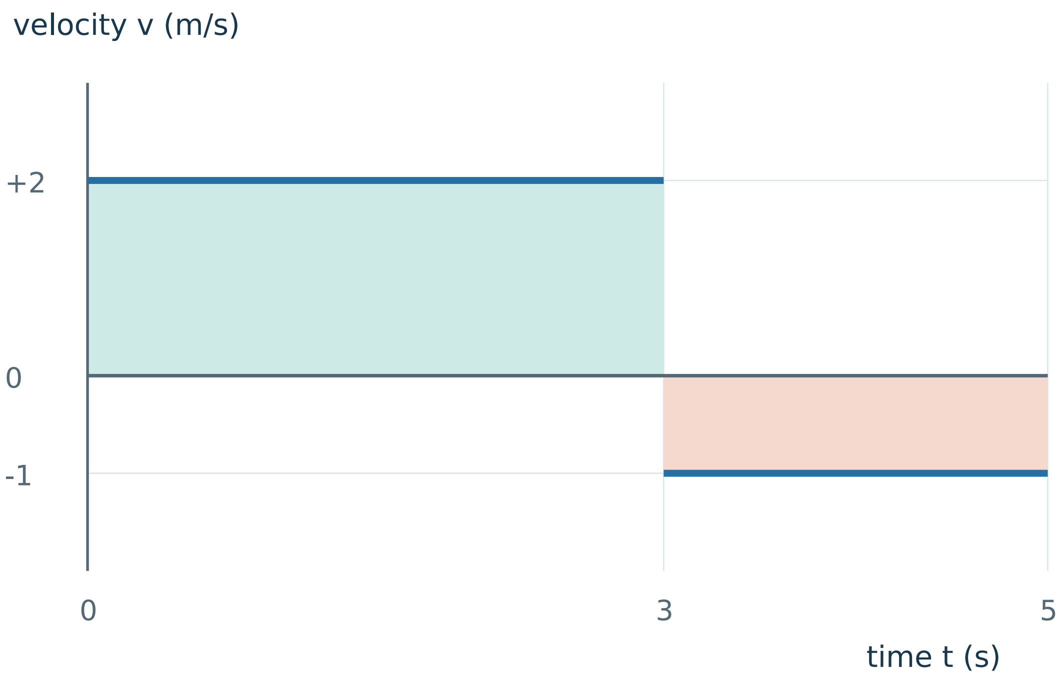
$$\text{Distance} = 12 + 2 = 14 \text{ m}$$

If you started at +5 m, you would finish at +15 m. The displacement would still be +10 m.

A return can cancel an earlier change in position. It cannot cancel distance already travelled. Complete the two areas in A6 yourself.

Your turn · Complete the area reasoning

The first region is worked for you. Complete the second region and combine them.



First region: (+2 m/s) × (3 s) = +6 m of rightward change.

G2a · Read the second width

The interval starts at 3 s and ends at 5 s.
Duration = 5 − 3 = ___ s.

G2b · Build its signed area

The velocity is −1 m/s.
Second change = (−1) × ___ = ___ m.

G2c · Keep two totals

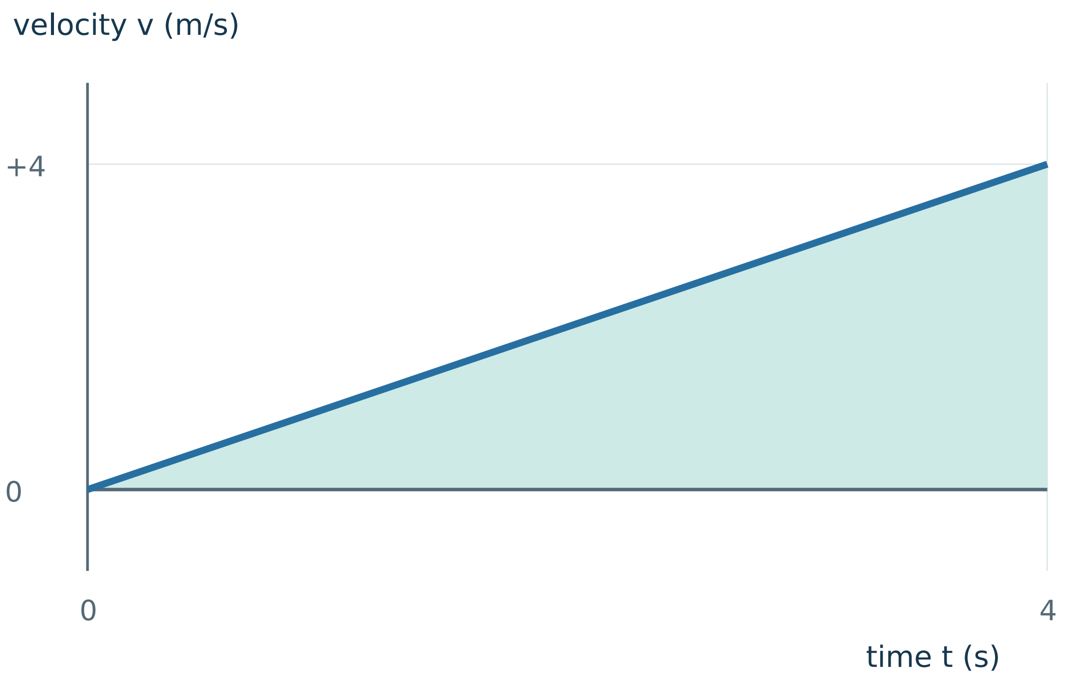
Displacement = +6 + (___) = ___ m.
Distance = 6 + ___ = ___ m.

If a sign is confusing, point to the graph region and say whether the object moves right or left. Then check the small-step solutions.

Check G2 · p. 17

When velocity rises steadily, use a triangle

Velocity rises in a straight line from 0 to +4 m/s over 4 s.



The sloping line cuts a 4 s by 4 m/s rectangle into two equal triangles.

Why not 4 × 4?

That would describe +4 m/s for all four seconds. Here the object starts at rest and builds up speed.

Take half of the rectangle

Area = ½ × 4 × 4 = 8 m

The ½ comes from the triangle, not from a special motion formula.

All the graph lies at or above zero. So displacement is +8 m and distance is 8 m.

Use ½ × base × height only when the region between the graph and zero line is a triangle.

Our triangle questions stay above zero. A straight segment away from zero may enclose a rectangle plus a triangle; do not call every sloping region a triangle.

Try A8 · p. 13

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A1 · Apply

Start at +1 m. Walk 5 m left, then 3 m right. Find the final position, distance and displacement.

Start → turning point → finish
Final position: ____
Distance: ____ Displacement: ____

Hints · p. 15 Solution · p. 18

A2 · Apply

An object moves from −2 m to +5 m without reversing. Find its distance and displacement. Explain the signs.

Sketch the line and explain why crossing zero is not a turn.

Hints · p. 15 Solution · p. 18

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A3 · Explain

Two students start at 0 m and finish at +4 m. Must they have travelled the same distance? Draw two journeys to support your answer.

Draw and label two routes. Keep the endpoints the same.

Hints · p. 15 Solution · p. 19

A4 · Apply

A student walks 7 m right and then 3 m left. Find distance and displacement, taking right as positive.

Show the route or working that makes your answer clear.

Hints · p. 15 Solution · p. 19

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A5 · Apply

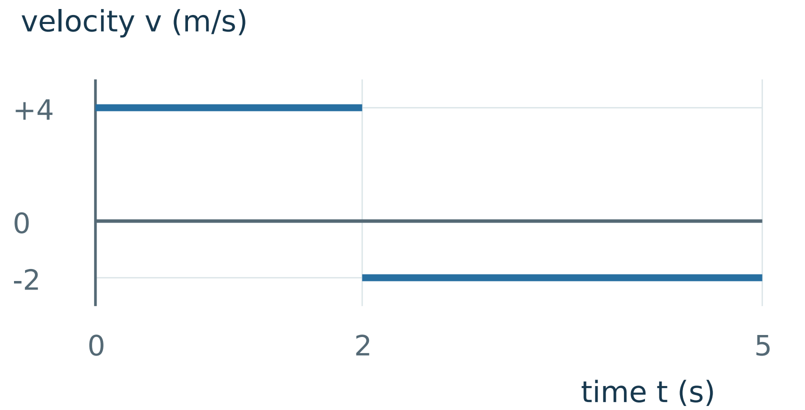
An object moves at +5 m/s for 3 s, then at −2 m/s for 2 s. Find displacement and distance for the whole journey.

Show the route or working that makes your answer clear.

Hints · p. 16 Solution · p. 20

A6 · Apply

Use the graph to find displacement and distance from 0 s to 5 s. Give a physical meaning for each area.



Hints · p. 16 Solution · p. 20

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A7 · Apply

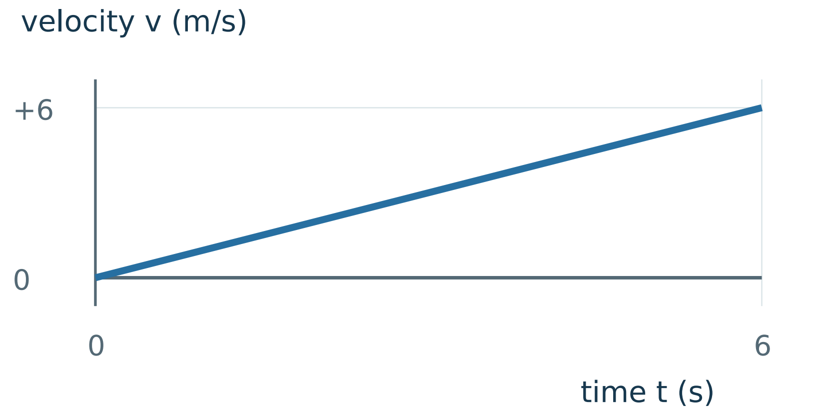
A constant velocity of +2.5 m/s is shown from 0 s to 8 s. Find displacement and distance. Why do their sizes agree?

Show the route or working that makes your answer clear.

Hints · p. 16 Solution · p. 21

A8 · Connect

A trolley has the velocity–time graph shown. Find displacement and distance from 0 s to 6 s. Explain why 6 × 6 is unsuitable.

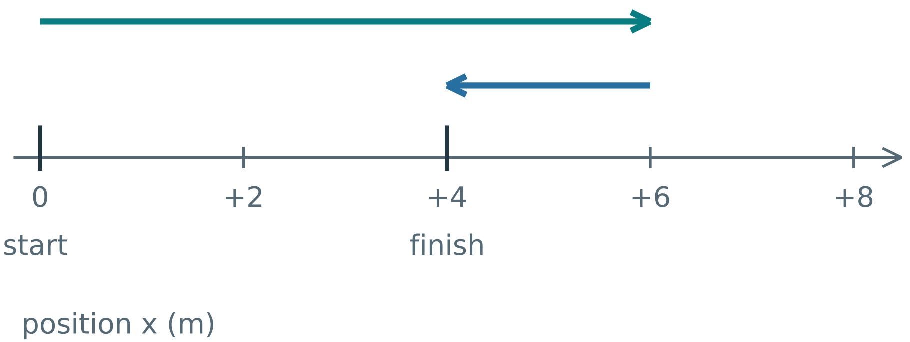


Hints · p. 16 Solution · p. 21

Appendix B · Keep the picture, rebuild the rule

Right is positive. Cover the words and explain each diagram aloud.

1 Distance and displacement



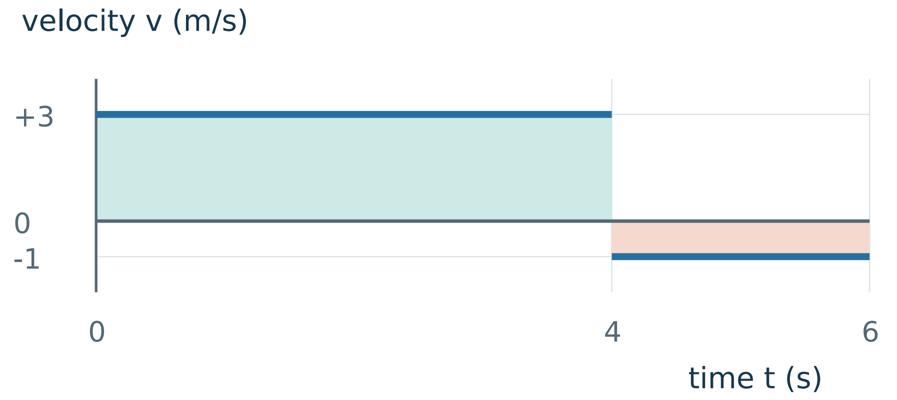
Distance: add every path length.
Displacement: the signed endpoint change.

$\Delta x = x_f - x_i$

Δx: displacement; x_i: initial position; x_f: final position. Here, distance is 8 m and Δx is +4 m.

Check: distance ≥ $|\Delta x|$. Bars mean size.
Rebuild: trace the path, then compare its endpoints.

2 Reading a velocity–time graph



Displacement: add signed areas.
Distance: add all the area sizes. Split at zero crossings.

Rectangle: v × duration

Triangle size: ½ × base × height.
Graph above: Δx = +10 m; distance 14 m.

Check: (m/s) × s = m. Use a rectangle for constant velocity. Use ½ × base × height only for a triangular region; other straight segments may need rectangle + triangle.

OPTIONAL HELP

Hints · Read one step, then try again

Cover the rows below the hint you are reading.

A1

H1 Notice Keep the starting coordinate separate from the lengths walked.

H2 Model Mark the start, the turning point and the finish on a number line.

H3 Start The turning point is at +1 − 5 m. Find the finish next.

Return · p. 10

A2

H1 Notice Crossing the origin does not mean reversing direction.

H2 Model Draw −2, 0 and +5 in order along the line.

H3 Start The first part of the path, from −2 m to 0 m, is 2 m.

Return · p. 10

A3

H1 Notice The question fixes the endpoints, not the route.

H2 Model Draw one direct route and one route with a turn.

H3 Start For the turning route, choose a turning point beyond +4 m.

Return · p. 11

A4

H1 Notice The student reverses direction after the first leg.

H2 Model Use a path line and a separate start-to-finish arrow.

H3 Start Add 7 m and 3 m to begin the distance calculation.

Return · p. 11

OPTIONAL HELP

Hints · Read one step, then try again

Cover the rows below the hint you are reading.

A5

H1 Notice One interval is rightward and one is leftward.

H2 Model Draw two velocity–time rectangles on opposite sides of zero.

H3 Start The first signed area is (+5 m/s) × (3 s).

Return · p. 12

A6

H1 Notice Read the duration of each horizontal segment.

H2 Model Separate the two rectangular regions at $t = 2$ s.

H3 Start For the first region, write $(+4) \times (2 - 0)$.

Return · p. 12

A7

H1 Notice The velocity stays on one side of zero.

H2 Model Draw the rectangle and label height and width with units.

H3 Start Write area = (+2.5 m/s) × (8 s).

Return · p. 13

A8

H1 Notice The height grows during the interval.

H2 Model Compare the region with the complete enclosing rectangle.

H3 Start The rectangle would have area (6 s) × (6 m/s); use the fraction actually occupied.

Return · p. 13

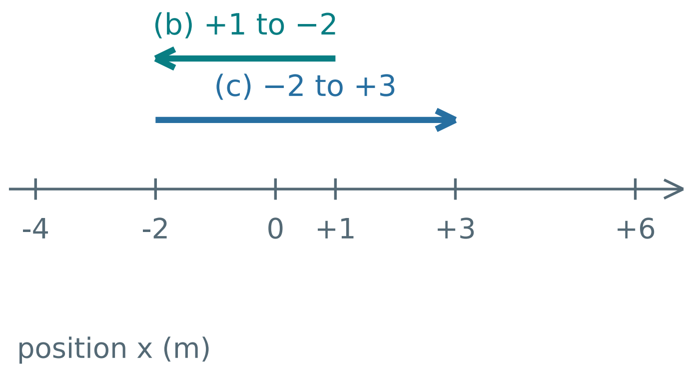
SOLUTIONS BEGIN HERE

Check the small steps

Use these pictures to repair a step before trying the complete problems.

G1 · Positions and one-step changes

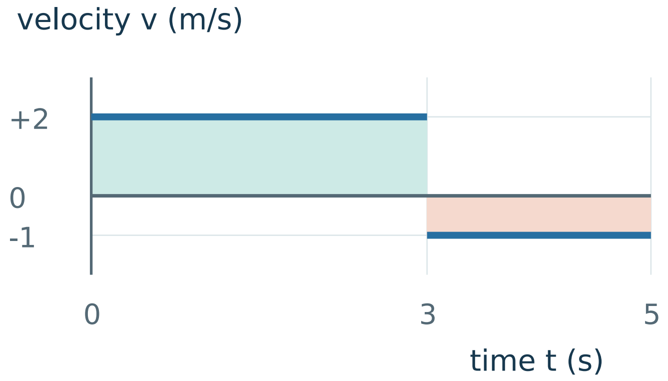
a. −2 m is 2 m left of the origin.
b. From +1 m, moving 3 m left ends at −2 m. The first arrow shows this move.
c. From −2 m to +3 m, move 2 m to zero, then 3 m right: displacement = +5 m (right). The second arrow shows this separate move.



Return · p. 2

G2 · Duration, signed change and total path

a. Duration = 5 − 3 = 2 s.
b. Second change = (−1) × 2 = −2 m: 2 m left.
c. Displacement = +6 + (−2) = +4 m. Distance = 6 + 2 = 8 m.
The leftward part reduces the endpoint change, but still adds 2 m to the path.



Return · p. 8

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

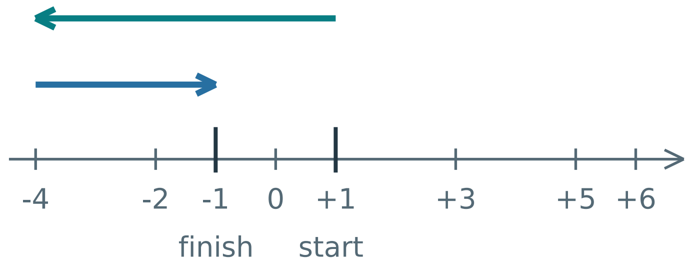
A1 Start +1 m; 5 m left, then 3 m right.

Why. Position changes carry direction; path lengths do not.

Method. Start +1 m → turn −4 m → finish −1 m. Distance = 5 + 3 = 8 m. Displacement = −1 − (+1) = −2 m.

Answer / check. Final position −1 m; distance 8 m; displacement −2 m (left).

Keep. The final coordinate is not automatically the displacement.



Return to A1 · p. 10

Lesson · p. 4

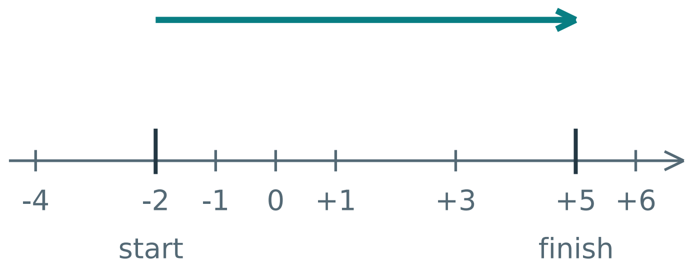
A2 Move from −2 m to +5 m with no reversal.

Why. The route goes 2 m to zero and another 5 m right.

Method. From −2 m to zero is 2 m; then 5 m more to +5 m. Total change is 7 m right. This explains +5 − (−2) = +7 m.

Answer / check. Distance 7 m; displacement +7 m (right).

Keep. Equal distance and displacement size require no reversal in this straight-line journey.



Return to A2 · p. 10

Lesson · p. 1

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A3 Same start 0 m and finish +4 m; compare possible distances.

Why. Endpoints fix displacement but leave the path undecided.

Method. One can go directly 0 → +4. Another can go 0 → +7 → +4. Their paths are 4 m and 7 + 3 m.

Answer / check. No. These distances are 4 m and 10 m; both displacements are +4 m.

Keep. A drawing can disprove an "always" claim.

Return to A3 · p. 11 Lesson · p. 3

A4 Walk 7 m right, then 3 m left.

Why. Both legs count towards distance, while leftward motion reduces the rightward change.

Method. Distance = 7 + 3. Displacement = +7 + (−3).

Answer / check. Distance 10 m; displacement +4 m (right).

Keep. Separate total path from net change.

Return to A4 · p. 11 Lesson · p. 3

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A5 Velocity +5 m/s for 3 s, then −2 m/s for 2 s.

Why. Each interval has constant velocity, so its signed area is velocity × duration.

Method. The changes are +5 × 3 = +15 m and −2 × 2 = −4 m. Add with signs for displacement; add sizes for distance.

Answer / check. Displacement +11 m; distance 19 m. Check: 19 ≥ 11.

Keep. Negative velocity gives negative displacement, not negative distance.

Return to A5 · p. 12 Lesson · p. 7

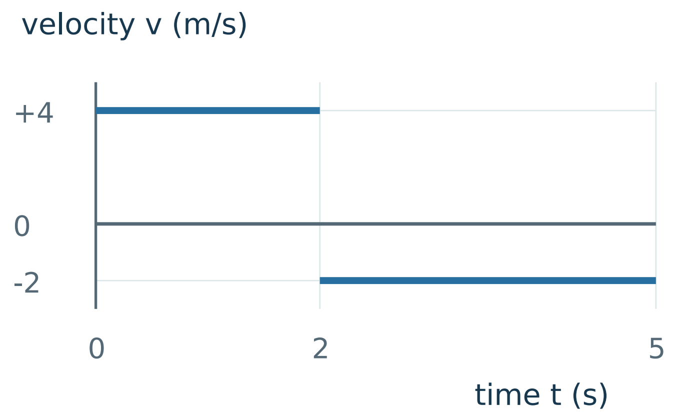
A6 Graph: +4 m/s from 0–2 s, then −2 m/s from 2–5 s.

Why. Graph widths are durations. The second width is 5 − 2 = 3 s.

Method. Signed areas are +4 × 2 = +8 m and −2 × 3 = −6 m. For distance count both path lengths.

Answer / check. Displacement +2 m; distance 14 m. The parts mean 8 m right and 6 m left.

Keep. Read each width from its endpoints before multiplying.



Return to A6 · p. 12 Lesson · p. 7

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A7 Constant velocity +2.5 m/s from 0–8 s.

Why. The graph is one positive rectangle; there is no reverse motion.

Method. Signed area = (+2.5 m/s) × (8 s) = +20 m. The same area size is the path length.

Answer / check. Displacement +20 m; distance 20 m. Their sizes agree because direction does not reverse.

Keep. A constant positive velocity produces steadily increasing position.

Return to A7 · p. 13 Lesson · p. 6

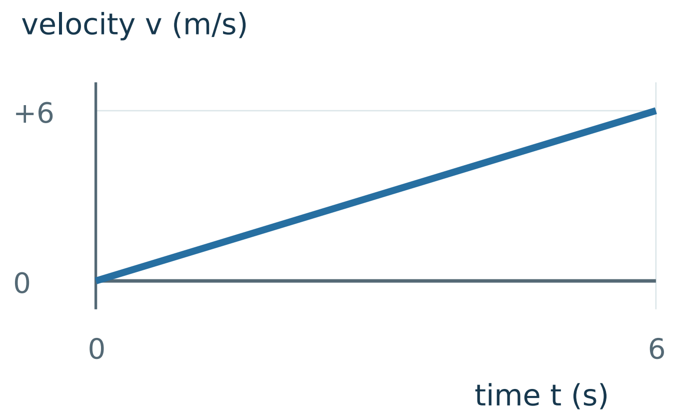
A8 Velocity increases linearly from 0 to +6 m/s in 6 s.

Why. The graph forms a triangle. The trolley does not spend all 6 s at its final velocity.

Method. The triangle is half of a 6 s by 6 m/s rectangle: ½ × 6 × 6 = 18 m.

Answer / check. Displacement +18 m; distance 18 m. The graph never goes below zero.

Keep. Choose the shape from the actual graph, not just its final height.



Return to A8 · p. 13 Lesson · p. 9